Guillermo Terrén-Serrano

Environmental Markets Lab (emLab), University of California, Santa Barbara

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Google Scholar

I am a computational power systems researcher working on the planning and operation of low-carbon electricity systems under uncertainty using optimization and probabilistic forecasting methods. My research integrates power system modeling, machine learning, energy meteorology, and electricity market analysis to study grid reliability, resource adequacy, and infrastructure investment under evolving climate and socioeconomic conditions. As a researcher and educator, I aim to mentor the next generation of leaders in intellectually ambitious and inclusive research environments that cultivate scientific curiosity, technical excellence, and collaborative problem-solving.

Research Interests — *Power systems optimization; stochastic optimization; resource adequacy and reliability; probabilistic forecasting; electricity markets; low-carbon electricity systems; uncertainty quantification; machine learning for energy systems; energy meteorology and remote sensing.*

Education

Ph.D. in Electrical Engineering (Pass with Distinction) 2016–2022
"Intra-hour Solar Forecasting using Cloud Dynamics Features Extracted from Ground-Based Infrared Sky Images." Department of Electrical & Computer Engineering, University of New Mexico, NM, USA. *Advisor: Prof. Martínez-Ramón*

M.Sc. in Power & Energy 2015–2016
"Machine Learning Approach for Global Solar Radiation Time-Series Forecasting." Department of Electrical & Computer Engineering, University of New Mexico, NM, USA. *Advisor: Prof. Martínez-Ramón*

B.Sc. in Electrical Engineering 2006–2011
Escuela de Ingeniería y Arquitectura, Universidad de Zaragoza, Zaragoza, Spain. *Advisor: Prof. Ibáñez-Carabantes*

Professional Experience

Postdoctoral Scholar 2022–Present
Environmental Markets Lab (emLab) and Environmental Studies Program, University of California, Santa Barbara, CA, USA.

Postdoctoral Research Associate 2022–2023
Department of Statistics and Applied Probability, University of California, Santa Barbara, CA, USA.

Graduate Research Assistant 2015–2022
Center for Emerging Energy Technologies (CEET) and Department of Electrical & Computer Engineering, University of New Mexico, NM, USA.

Project Engineer 2012–2013
Photovoltaic power plants, Enerland Group, Zaragoza, Spain.

Undergraduate Research Assistant 2012
Electronic Engineering Department, Institute of Technology for the Development (LACTEC), PR, Brazil.

Peer-Reviewed Publications

(†indicates equal contribution) citations=491, h-index=10, i10-index=11

13. **G. Terrén-Serrano**, R. Deshmukh and M. Martínez-Ramón. Probabilistic Day-Ahead Forecasting of System-Level Renewable Energy and Electricity Demand. *Nature Communications*, 2026.
12. **G. Terrén-Serrano** and M. Ludkovski. Extreme Day-Ahead Renewables Scenario Selection in Power Grid Operations. *Applied Energy*, 2025.
11. Q. Paletta, **G. Terrén-Serrano**, Y. Nie, B. Li, J. Bieker, W. Zhang, L. Dubus, S. Dev and C. Feng. Advances in Solar Forecasting: Computer Vision with Deep Learning. *Advances in Applied Energy*, 2023.
10. **G. Terrén-Serrano** and M. Martínez-Ramón. Deep Learning for Intra-Hour Solar Forecasting with Fusion of Features Extracted from Infrared Sky Images. *Information Fusion*, 2023.
9. **G. Terrén-Serrano** and M. Martínez-Ramón. Detection of Multiple Wind Velocity Fields using Ground-Based Infrared Sky Images. *Knowledge-Based Systems*, 2023.
8. **G. Terrén-Serrano** and M. Martínez-Ramón. Kernel Learning for Intra-Hour Solar Forecasting with Infrared Sky Images and Cloud Dynamic Feature Extraction. *Renewable and Sustainable Energy Reviews*, 2023.
7. **G. Terrén-Serrano** and M. Martínez-Ramón. Processing of Global Solar Irradiance and Ground-Based Infrared Sky Images for Solar Nowcasting and Intra-Hour Forecasting Applications. *Solar Energy*, 2023.
6. **G. Terrén-Serrano** and M. Martínez-Ramón. Geospatial Perspective Reprojections for Ground-Based Sky Imaging Systems. *IEEE Transactions on Geoscience and Remote Sensing*, 2022.
5. **G. Terrén-Serrano** and M. Martínez-Ramón. Comparative Analysis of Methods for Cloud Segmentation in Ground-Based Infrared Images. *Renewable Energy*, 2021.
4. **G. Terrén-Serrano** and M. Martínez-Ramón. Multi-Layer Wind Velocity Field Visualization in Infrared Images of Clouds for Solar Irradiance Forecasting. *Applied Energy*, 2021.
3. **G. Terrén-Serrano**, A. Bashir, T. Estrada and M. Martínez-Ramón. Girasol, a Sky Imaging and Global Solar Irradiance Dataset. *Data in Brief*, 2021.
2. A. Mammoli, **G. Terrén-Serrano**, A. Menicucci, T.P. Caudell and M. Martínez-Ramón. An Experimental Method to Merge Far-Field Images from Multiple Long-Wave Infrared Sensors for Short-Term Solar Forecasting. *Solar Energy*, 2019.
1. O. García-Hinde, **G. Terrén-Serrano**, M.Á. Hombrados-Herrera, V. Gómez-Verdejo, S. Jiménez-Fernández, C. Casanova-Mateo, J. Sanz-Justo, M. Martínez-Ramón and S. Salcedo-Sanz. Evaluation of Dimensionality Reduction Methods Applied to Numerical Weather Models for Solar Radiation Forecasting. *Engineering Applications of Artificial Intelligence*, 2018.

Manuscripts Under Review

G. Terrén-Serrano and M. Ludkovski. Dynamic Intraday Probabilistic Forecasting Update of Renewable Power Generation. *Submitted*, 2026.

G. Terrén-Serrano†, R. Deshmukh†, S. Dukkipati, M. Meng, S. Prasad, A. Mileva and A. Gambhir. Affordable Low-Carbon Electricity Pathways for India under Uncertainty. *Submitted* (preprint on arXiv), 2026.

Manuscripts in Preparation

G. Terrén-Serrano, A. Sharma, R. Deshmukh and S. Prasad. Resource Adequacy in Electricity Systems under Demand Growth and Decarbonization. *In preparation.*

G. Terrén-Serrano and R. Deshmukh. Reliability Contribution of Renewable Energy Resources in Large Grids under Decarbonization. *In preparation.*

W. T. Wang, S. Prasad, **G. Terrén-Serrano** and R. Deshmukh. Employment, Health, and Environmental Effects of India's Power System Low-Carbon Transition. *In preparation.*

Conference Proceedings

G. Terrén-Serrano, R. Deshmukh and M. Martínez-Ramón. Day-Ahead Operational Forecast of Aggregated Solar Generation Assimilating Mesoscale Meteorology Information. *IEEE PES Grid Edge Technologies Conference & Exposition*, San Diego, USA, 2025.

G. Terrén-Serrano and M. Martínez-Ramón. Explicit Basis Function Kernel Methods for Cloud Segmentation in Infrared Images. *Energy Reports*, 2022.

G. Terrén-Serrano and M. Martínez-Ramón. Wind Flow Estimation in Thermal Sky Images for Sun Occlusion Prediction. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022.

G. Terrén-Serrano and M. Martínez-Ramón. Segmentation Algorithms for Ground-Based Infrared Cloud Images. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022.

Grants and Funding

(*indicates equivalent; postdoctoral scholars at UCSB are not eligible to serve as PI or co-PI)

Resource Adequacy and Long-Term Planning of India's Electricity System. *World Resources Institute (WRI)*, 2024–2026. PI: R. Deshmukh. Role: Investigator. Total: $450,000.

CNSI Climate Innovation Fund. *California NanoSystems Institute*, 2023. Computing resources and conference travel. PI: R. Deshmukh. Role: Co-PI*. Total: $30,000.

Research Seed Program. *UCSB Institute for Energy Efficiency*, 2022. Software development and conference travel. PI: R. Deshmukh. Role: Co-PI*. Total: $50,000.

Stochastic Models, Indices & Optimization Algorithms for Pricing & Hedging Reliability Risks in Modern Power Grids. *ARPA-E PERFORM*, 2022–2023. PI: R. Carmona. Role: Investigator. Total: $3,500,000.

The New Mexico SMART Grid Center: Sustainable, Modular, Adaptive, Resilient, and Transactive. *NSF EPSCoR*, 2018–2025. PI: G. Balakrishnan. Role: Investigator. Total: $20,000,000.

Honors and Awards

2023–2025 CNSI Climate Innovation Postdoctoral Fellowship. *California NanoSystems Institute.* Competitive fellowship to research technologies for California's climate goals. Supported: extension of the joint day-ahead forecasting from solar generation (2022 Research Seed Program) to wind and electricity demand.

2015–2022 UNM-Avangrid Graduate Scholarship. *Avangrid (Iberdrola) Foundation.* Competitive scholarship to complete doctoral studies in the ECE department at UNM under the King Felipe VI of Spain Endowed Chair.

2011 UZ-Santander Americampus Scholarship. *Santander Bank.* Competitive scholarship to complete senior project design on an academic exchange year at the PUC-PR, Curitiba, Brazil.

Teaching and Mentoring Experience

Instructor of record:

ES 105 Renewable Energy Systems. Environmental Studies Program, UCSB. Fall 2024.

Teaching assistant:

ECE 213 Circuit Analysis II. Electrical and Computer Engineering (ECE), UNM.

ECE 203 Circuit Analysis I. ECE, UNM.

ENG 302 Fundamentals of Engineering: Electronic Circuits. School of Engineering, UNM.

ECE 314 Signals and Systems. ECE, UNM.

ECE 331 Data Structures and Algorithms. ECE, UNM.

Guest lecturer:

ECE 417 Advanced Machine Learning. Probabilistic Principal Component Analysis. ECE, UNM.

Mentor:

Bryan Lu. Research internship, Summer 2026, UCSB.

Steven De Jesus. EUREKA! Program, Summer 2023, UCSB.

David Lopez. EUREKA! Program + CAMP Research School Year Internship, 2022–2023, UCSB.

Alexander Santos-Lozada. CIESESE Program, Summers 2017 and 2018, UNM.

Invited Talks and Conference Presentations

2025 Forecasting & Markets Workshop. *Presentation.* ESIG, TN, USA.

2025 Institute for Energy Efficiency (IEE). *Seminar.* UCSB, CA, USA.

2024 ARPA-E Energy Innovation Summit. *Exhibition Booth.* Department of Energy, TX, USA.

2024 Macro-Energy Systems (MES) Workshop. *Flash Talk + Poster.* Princeton University, NJ, USA.

2024 MIT Energy Initiative. *Seminar.* Massachusetts Institute of Technology, MA, USA.

2023 International Energy Workshop (IEW). *Presentation.* Colorado School of Mines, CO, USA.

Datasets, Models, and Packages

G. Terrén-Serrano, R. Deshmukh, S. Dukkipati, M. Meng, S. Prasad, A. Mileva and A. Gambhir. *GridPath-India long-term (2020–2050) power system planning model data.* Dryad, (2026). Dataset & Model.

G. Terrén-Serrano, S. Prasad and R. Deshmukh. *GridPath-India workshops: illustrative models for Rajasthan, Madhya Pradesh, and Tamil Nadu.* GitHub, (2025). Model.

G. Terrén-Serrano, A. Bashir, T. Estrada and M. Martínez-Ramón. *Girasol, a sky imaging and global solar irradiance dataset.* Dryad, (2021). Dataset.

Institutional Service and Community Involvement

Reviewer. Applied Energy; Renewable Energy; Knowledge-Based Systems; Solar Energy; Data in Brief; Wiley (book proposal).

Student Athlete. Rugby team, Universidad de Zaragoza, Zaragoza, Spain (2006–2011).

Spanish Translator. Parent-teacher conferences, South Valley Academy, NM, USA (2020).

Skills

OS: Linux, Ubuntu, macOS, Windows, Raspbian.

Programming Languages: Python, R, Matlab, C/C++, Bash, LabVIEW, Simulink, HTML, Pascal.

Scientific Computing: CPU, GPU, and MPI parallel computing.

Prototyping: Arduino and Raspberry Pi, and 3D printing (CAD).

Modules: TensorFlow, Keras, PyTorch, GridPath, PyPSA.

Writing: LaTeX, Overleaf, Microsoft Office, Google Docs.

Version Control: Git, GitHub.

Professional References

Prof. Ranjit Deshmukh (rdeshmukh@ucsb.edu). *Associate Professor.* Bren School of Environmental Science & Management, and Environmental Studies Program, UCSB, CA, USA.

Prof. Manel Martínez-Ramón (manel@unm.edu). *Professor.* Department of Electrical and Computer Engineering, UNM, NM, USA.

Prof. Michael Ludkovski (mludkov@ucsb.edu). *Professor.* Department of Statistics and Applied Probability, UCSB, CA, USA.

Dr. Andrea Mammoli (aamammo@sandia.gov). *Principal Member of Technical Staff.* Sandia National Laboratories, NM, USA.